EDUARDO PIGNATELLI

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EDUCATION

1. **PhD in Reinforcement Learning**
   University College London
   November 2020 to 2024
   London, United Kingdom
   Subject: Credit Assignment in Deep Reinforcement Learning

2. **Master of Science in Architecture**
   University of Naples Federico II
   2015
   Naples, Italy
   Subject: Architecture and Engineering

EXPERIENCE

1. **Assistant Professor (UK Lecturer, Teaching)**
   University College London
   September 2024 to **present**
   London, UK

2. **Research Assistant**
   Imperial College London
   September 2019 to November 2020
   London, UK

3. **Machine Learning and Decision Analytics Lead**
   BuroHappold Engineering
   August 2018 to October 2020
   London, UK

4. **Computational/Machine Learning Engineer**
   BuroHappold Engineering
   August 2017 to August 2018
   London, UK

5. **Intern Computational Engineer**
   BuroHappold Engineering
   April 2017 to August 2017
   Bath, UK

6. **Architect**
   Gianni Ranaulo Design
   September 2016 to December 2016.
   Dubai, UAE

7. **Computational Architect**
   Gridshell.it
   September 2015 to September 2016
   Naples, Italy

8. **Computational Architectural Assistant**
   Gridshell.it
   July 2014 to September 2015
   Naples, Italy

9. **Intern Architect**
   CRC – Constructions Restorations and Consolidations
   September 2012 to December 2012
   Naples, Italy

CONSULTING

1. **Machine Learning consultant**
   University College London Consultants for **Adler Ortho**
   December 2025 to present
   London, UK
   Machine Learning for the generation of customised hip implants.
2. **Teaching consultant**
   Imperial College for **Sberbank**
   March 2020 to March 2021
   London, UK
   Class: **Computer Vision and Image retrieval.**
3. **Teaching Assistant**
   University College London
   March 2020 to present
   London, UK
   Classes:
   - **UCL-COMP0089**: *Reinforcement Learning*
   - **UCL-ELEC0136**: *Data Acquisition and Processing Systems*
4. **Teaching consultant**
   University College London Consultants for **Defence Science and Technology Laboratory**
   January 2021 to present
   London, UK
   Classes:
   - **Machine Learning**
   - **Deep Reinforcement Learning**

PUBLICATIONS

1. Paglieri D., Cupiał B., Coward S., Piterbarg U., Wolczyk M., Khan A., **Pignatelli E.**, Kuciński Ł., Pinto L., Fergus R., Foerster J. N., Parker-Holder J., Rocktäschel T., 2025. *AgentQuest: Benchmarking LLM and VLM Agents on Long-Horizon Interactive Tasks*. **ICLR** 2025.
2. **Pignatelli, E.**, Liesen, J., Lange, R.T., Lu, C., Castro, P.S. and Toni, L., 2024. *NAVIX: Scaling MiniGrid Environments with JAX*. In review at **NeurIPS**.
3. **Pignatelli, E.**, Ferret, J., Paglieri, D., Coward, S., Tim Rocktäschel, Edward Grefenstette, Toni, L., 2024. *Assessing the Zero-Shot Capabilities of LLMs for Action Evaluation in RL*. **ICML** Workshop on Auto Reinforcement Learning
4. **Pignatelli, E.**, Ferret, J., Geist, M., Mesnard, T., van Hasselt, H., Toni, L., 2024. *A Survey of Temporal Credit Assignment in Deep Reinforcement Learning*. **TMLR**. **Awarded Survey Certification.**
5. A. Kayal, **Pignatelli, E.**, Toni, L., 2023. *Does Behavioural Diversity in Intrinsic Rewards help Exploration?*. In **NeurIPS** Second Agent Learning in Open-Endedness Workshop.
6. Ntagiantas, K., **Pignatelli, E.**, Peters, N. S., Cantwell, C. D., Chowdhury, R. A., Bharath, A. A., 2023. *Estimation of Fibre Architecture and Scar in Myocardial Tissue Using Electrograms: an in-silico study*. In *Biomedical Signal Processing and Control*.
7. Wong N., Meshkinfamfard S., Turbé V., Whitaker M., Moshe M., Bardanzellu A., Dai T., **Pignatelli E.**, Barclay W., Darzi A., Elliott P., Ward H., Tanaka R., Cooke G., McKendry R., Atchison C., Bharath A., 2022. *Machine learning to support visual auditing of home-based lateral flow immunoassay self-test results for SARS-CoV-2 antibodies*. **Communications Medicine. Nature Research**.
8. Lino, M., Cantwell, C., Fotiadis, S., **Pignatelli, E.**, Bharath, A., 2020. *Simulating Surface Wave Dynamics with Convolutional Networks*. In **NeurIPS** workshop on Interpretable Inductive Biases and Physically Structured Learning.
9. Fotiadis, S., **Pignatelli, E.**, Valencia, M.L., Cantwell, C., Storkey, A., Bharath, A.A., 2020. *Comparing recurrent and convolutional neural networks for predicting wave propagation*. In **ICLR** Workshop on Deep Neural Models and Differential Equations.
10. Di Rosario, **S., Pignatelli**, E. and Mirra, G., 2018, May. *An automated design methodology for acoustic shells in outdoor concerts*. In Proceedings of the **EuroNoise** (Vol. 2018, pp. 2123-2130).
11. **Pignatelli, E.**, Mirra, G. and Pone, S., 2017, September. *InFormer. designing forming actions in post-formed gridshells by means of Multi-Objective Genetic Algorithms*. In Proceedings of **IASS** Annual Symposia (Vol. 2017, No. 17, pp. 1-10). International Association for Shell and Spatial Structures (IASS).
12. **Pignatelli, E.**, Mirra, G., and Pone, S., 2016, September. *Computational morphogenesis and construction of an acoustic shell for outdoor chamber music*. In Proceedings of **IASS** Annual Symposia (Vol. 2016, No. 17, pp. 1-10). International Association for Shell and Spatial Structures (IASS).
13. Pone, S., Mirra, G., **Pignatelli, E.**, Lancia, D. and Colabella, S., 2016, October. *Specialised algorithms for different project stages in a post-formed timber gridshell design*. In Proceedings of the 3rd International Conference on Structures and Architecture **(ICSA)** (pp. 259-266).
14. Di Rosario, S., Parenti, B., **Pignatelli, E.**, Mirra, G., Pone, S., 2015, October. *Res, Resonant String Shell, development and design of an acoustic shell for outdoor chamber music concerts*. In Proceedings of the **Institute of Acoustics** (Vol. 37, pp. 354-373). 9th International Conference on Auditorium Acoustics.
15. **Pignatelli, E.**, Colabella, S., Rosario, S.D. and Pone, S., 2015, August. *A wooden acoustic shell for open-air chamber music concert*. In Proceedings of **IASS** Annual Symposia (Vol. 2015, No. 25, pp. 1-12). International Association for Shell and Spatial Structures (IASS).

AWARDS AND SCHOLARSHIPS

1. Nomination for “UCL Inspiring Teaching Delivery Award”
2. TMLR Survey certification for “A survey of Temporal Credit Assignment in Deep Reinforcement Learning”
3. UK Research and Innovation Studentship
4. Best Innovation 2020 – Society of Digital Engineering
5. Peter Lord Award
6. Honorary Fellowship: Authority in the subject (Cultore della Materia)
7. Essence of Buro Happold Award
8. M.A.R Scholarship – https://www.vpmusica.com/